

Causal Inference: Retention Offer Use Case

Chapter 12: Inverse Probability Weighting & Marginal Structural Models

Applied Causal Inference Series — Use Case: “Did the Retention Offer Actually Reduce Churn?”

Executive Summary

Part I of *What If* (Chapters 1–10) introduced Inverse Probability (IP) weighting as a non-parametric method to create a pseudo-population in which treatment A is conditionally independent of baseline confounders L . However, non-parametric IP weighting fails in realistic CRM settings when L is high-dimensional or continuous. Chapter 12 details how **parametric propensity score models** (e.g., logistic regression) enable IP weighting for complex data. Furthermore, it introduces **Marginal Structural Models (MSMs)** for counterfactual outcomes, **stabilized IP weights** (SW^A) to reduce sampling variance, and **joint IP weights** ($W^{A,C}$) to adjust for loss to follow-up / missing data.

1 12.1 The Causal Question & CRM Context

(Hernán & Robins, Section 12.1, pp. 157–158)

Evaluating a customer retention program requires estimating the average causal effect of a retention offer ($A = 1$: offer sent, $A = 0$: no offer) on customer churn ($Y = 1$: churned, $Y = 0$: retained).

The naive associational risk difference from CRM database records:

$$\text{Associational Risk Difference} = \Pr[Y = 1 \mid A = 1] - \Pr[Y = 1 \mid A = 0]$$

suffers from severe **confounding bias**. High-risk subscribers (e.g., low monthly usage, high customer service tickets, short tenure) are selectively targeted for retention offers. To isolate the true **Average Causal Effect (ACE)**:

$$\text{ACE} = E[Y^{a=1}] - E[Y^{a=0}]$$

we must adjust for the baseline covariate vector $L = (L_1, L_2, \dots, L_p)$ comprising tenure, spend, product usage, and support history.

2 12.2 Estimating IP Weights via Modeling

(Hernán & Robins, Section 12.2, pp. 158–161)

2.1 The Pseudo-Population Concept

IP weighting assigns each subscriber a weight equal to the reciprocal of the conditional probability of receiving their actual treatment. This creates a **pseudo-population** in which:

1. Confounders L no longer predict treatment A ($L \perp\!\!\!\perp A$ in the pseudo-population).
2. Under conditional exchangeability ($Y^a \perp\!\!\!\perp A \mid L$), observational associations in the pseudo-population equal causal contrasts in the target population: $E_{\text{ps}}[Y \mid A = a] = E[Y^a]$.

2.2 Parametric Propensity Score Estimation

Non-stabilized IP weights are defined as:

$$W_i^A = \frac{1}{f(A_i | L_i)} = \begin{cases} \frac{1}{\Pr[A = 1 | L_i]} & \text{if } A_i = 1 \\ \frac{1}{1 - \Pr[A = 1 | L_i]} & \text{if } A_i = 0 \end{cases}$$

When L contains continuous variables (e.g., tenure in months, data volume), empirical cell counts yield 0/0. We resolve this by fitting a parametric logistic regression model:

$$\text{logit } \Pr[A = 1 | L] = \alpha_0 + \alpha_1 L_1 + \alpha_2 L_2 + \cdots + \alpha_p L_p$$

2.3 Estimating the Causal Effect

Using predicted propensity scores $\widehat{\Pr}[A = 1 | L]$, we fit a weighted linear mean model via Weighted Least Squares (WLS):

$$E[Y | A] = \theta_0 + \theta_1 A$$

weighted by $\widehat{W}^A$. The parameter estimate $\widehat{\theta}_1$ is an unbiased point estimate of the causal risk difference $E[Y^{a=1}] - E[Y^{a=0}]$. Robust (sandwich) standard errors or non-parametric bootstrapping are required for valid 95% confidence intervals.

3 12.3 Stabilized IP Weights

(Hernán & Robins, Section 12.3, pp. 161–163)

Non-stabilized weights $W^A = \frac{1}{f(A|L)}$ can take extreme values when $\Pr[A = 1 | L] \approx 0$ or $\Pr[A = 1 | L] \approx 1$. Extreme weights produce high sampling variability and wide confidence intervals.

To stabilize weights, we multiply the numerator by the marginal probability of treatment $f(A)$:

$$SW^A = \frac{f(A)}{f(A | L)} = \begin{cases} \frac{\Pr[A = 1]}{\Pr[A = 1 | L]} & \text{if } A = 1 \\ \frac{\Pr[A = 0]}{1 - \Pr[A = 1 | L]} & \text{if } A = 0 \end{cases}$$

Table 1: Comparison of Non-Stabilized and Stabilized IP Weights

Weight Type	Formula	Mean	Typical Range	Sampling Variance
Non-Stabilized (W^A)	$\frac{1}{f(A L)}$	≈ 2.0	1.05 – 16.7	High (Wider CIs)
Stabilized (SW^A)	$\frac{f(A)}{f(A L)}$	≈ 1.0	0.33 – 4.30	Low (Narrower CIs)

4 12.4 Marginal Structural Models (MSMs)

(Hernán & Robins, Section 12.4, pp. 163–165)

A **Marginal Structural Model (MSM)** specifies a functional form for the expected value of a counterfactual outcome Y^a :

$$E[Y^a] = \beta_0 + \beta_1 a$$

It is *marginal* because it models target population counterfactual means without conditioning on L , and *structural* because its outcome is a counterfactual variable Y^a .

4.1 Continuous & Multi-Valued Treatments

When A is a continuous treatment (e.g., continuous discount percentage $A \in [0\%, 50\%]$), we specify a non-saturated dose-response MSM:

$$E[Y^a] = \beta_0 + \beta_1 a + \beta_2 a^2$$

We fit the weighted outcome model $E[Y | A] = \theta_0 + \theta_1 A + \theta_2 A^2$ in the pseudo-population weighted by SW^A . For continuous A , $f(A | L)$ is a conditional probability density function (PDF), typically modeled as $A | L \sim \mathcal{N}(\mu_L, \sigma^2)$.

5 12.5 Effect Modification in MSMs

(Hernán & Robins, Section 12.5, pp. 165–166)

To evaluate whether treatment effects differ across baseline customer segments (e.g., contract type V), we include V in the MSM:

$$E[Y^a | V] = \beta_0 + \beta_1 a + \beta_2 (a \times V) + \beta_3 V$$

Parameters are estimated using modified stabilized weights that condition on V in the numerator:

$$SW^A(V) = \frac{f(A | V)}{f(A | L)}$$

6 12.6 Censoring and Missing Data

(Hernán & Robins, Section 12.6, pp. 166–168)

When subscribers drop out or have unrecorded status ($C = 1$: censored, $C = 0$: uncensored), restricting analysis to $C = 0$ introduces **selection bias**.

To simultaneously adjust for confounding and selection bias, we construct joint IP weights:

$$W^{A,C} = W^A \times W^C = \frac{1}{f(A | L)} \times \frac{1}{\Pr[C = 0 | A, L]}$$

Joint stabilized weights take the form:

$$SW^{A,C} = SW^A \times SW^C = \frac{f(A)}{f(A | L)} \times \frac{\Pr[C = 0 | A]}{\Pr[C = 0 | A, L]}$$

Summary of Identifiability Conditions for MSMs

1. **Conditional Exchangeability:** $Y^{a,c=0} \perp\!\!\!\perp (A, C) | L$ (no unmeasured confounding or selection bias).
2. **Positivity:** $\Pr[A = a, C = 0 | L = l] > 0$ for all strata l where $\Pr[L = l] > 0$.
3. **Consistency:** $Y^{A,C=0} = Y$ for observed uncensored units under well-defined interventions.
4. **Correct Model Specification:** Correct functional forms for treatment propensity $f(A | L)$, censoring probability $\Pr[C = 0 | A, L]$, and structural model $E[Y^a]$.